



Barker College

**2008
TRIAL
HIGHER SCHOOL
CERTIFICATE**

Mathematics Extension 2

Staff Involved:

PM MONDAY 11 AUGUST

- BHC*
- BTP*
- JM
- WMD

35 copies

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen
- Make sure your Barker Student Number is on ALL pages of your answer sheets
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper

Total marks – 120

- Attempt Questions 1–8
- All questions are of equal value
- ALL necessary working should be shown in every question
- Start each question on a NEW page
- Write on one side of each answer page
- Marks may be deducted for careless or badly arranged work

Answer each question on a SEPARATE sheet of paper

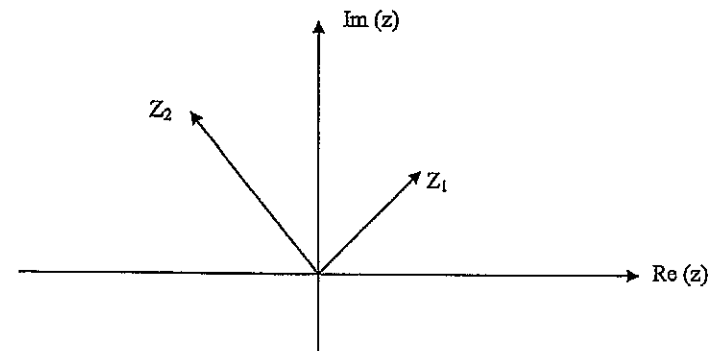
Question 1 (15 marks) [START A NEW PAGE] Marks

- (a) If $(\sqrt{3} + i)z = 4\sqrt{3} - 4i$, find
- (i) z in the form $a + bi$ 1
 - (ii) $|z|$ 1
 - (iii) $\arg(z)$ 1
 - (iv) z^3 in the form $a + bi$ 1
- (b) Find $\sqrt{-5 - 12i}$ in the form $a + bi$, and hence solve the equation $z^2 + (1 - 2i)z + (\frac{1}{2} + 2i) = 0$ 4
- (c) If w is a complex cube root of unity, show that $1 + w + w^2 = 0$, and hence prove that $(1 + w)(1 + 2w)(1 + 3w)(1 + 7w) = 31 + 2w$. 3

Question 1 continues on page 3

Question 1 (continued)

- (d) Let z_1 and z_2 be two given complex numbers as shown on the Argand diagram below.



Let z be a variable complex number.

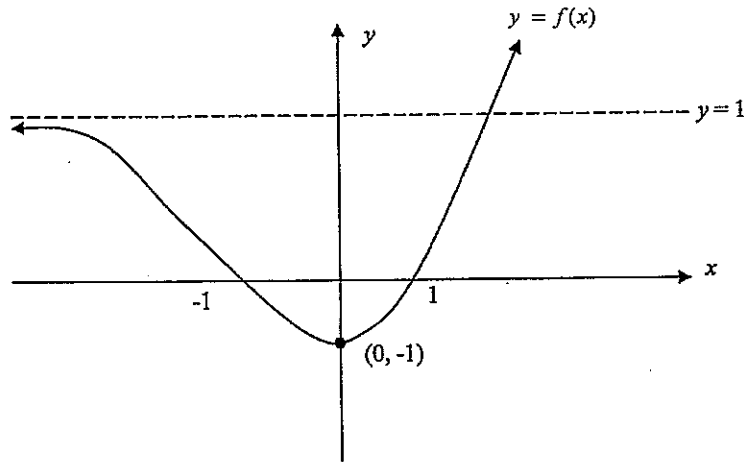
Sketch and describe the locus of z on an Argand diagram if:

- (i) $|z - z_1| = |z - z_2|$ 2
- (ii) $\arg\left(\frac{z - z_1}{z - z_2}\right) = \alpha$, where $0 < \alpha < \pi$ 2

End of Question 1

Question 2 (15 marks) [START A NEW PAGE]

- (a) The diagram below shows the graph of
- $y = f(x)$
- .

There is a minimum turning point at $(0, -1)$.

On separate diagrams, draw the graph of

- | | | |
|-------|-----------------------|---|
| (i) | $y = f(x)$ | 2 |
| (ii) | $y^2 = f(x)$ | 2 |
| (iii) | $y = \sin^{-1}[f(x)]$ | 2 |
| (iv) | $y = \ln[f(x)]$ | 2 |

Question 2 (continued)

- b) The equation of a curve is given by
- $xy^2 + x^2 = 1$

- | | | |
|-------|--|---|
| (i) | Explain why $x = 0$ is not in the domain of the curve. | 1 |
| (ii) | Find the x -intercepts of the curve. | 1 |
| (iii) | Re-write the equation of the curve making y the subject, and hence find the domain of the curve. | 2 |
| (iv) | The curve has two asymptotes. Write down the equations of both asymptotes. | 2 |
| (v) | Hence sketch the curve $xy^2 + x^2 = 1$ | 1 |

End of Question 2

Marks

Question 3 (15 marks) [START A NEW PAGE]

(a) Find

(i) $\int \frac{(x+3) dx}{\sqrt[3]{x^2+6x}}$ using the substitution $u = x^2 + 6x$ 2

(ii) $\int \frac{dy}{y^2 + 10y + 30}$ 2

(iii) $\int \frac{dx}{2 + \cos x}$ using the “t-results” 3

(iv) $\int x^2 e^{2x} dx$ 3

(b) Factorise $x^3 + x^2 - 6x$ and then find the values of A , B and C such that

$$\frac{x+1}{x^3+x^2-6x} = \frac{A}{x} + \frac{B}{x-2} + \frac{C}{x+3}$$

Hence find $\int \frac{(x+1) dx}{x^3+x^2-6x}$ 5

End of Question 3

Marks

Question 4 (15 marks) [START A NEW PAGE]

(a) Let $I_n = \int_0^{\pi/4} \tan^n x dx$

Show that $I_n + I_{n-2} = \frac{1}{n-1}$

Deduce the value of I_5 . 5

(b) Find the volume of the torus generated by revolving the circle $x^2 + y^2 = 4$ about the line $x = 3$. 4

(c) (i) Show that

$$\frac{d}{dx} \left\{ \frac{1}{2} x \sqrt{a^2 - x^2} + \frac{1}{2} a^2 \sin^{-1} \left(\frac{x}{a} \right) \right\} = \sqrt{a^2 - x^2}$$
 3

(ii) The base of a solid is the circle $x^2 + y^2 = 16x$. Every slice of this solid taken perpendicular to the x axis is a rectangle of height 6 units. Using the result from part (i) above, find the volume of this solid. 3

End of Question 4

Question 5 (15 marks) [START A NEW PAGE]

- (a) Consider the polynomial

$$p(x) = ax^4 + bx^3 + cx^2 + d$$

where a , b , c and d are integers.

Suppose that α is an integer such that $p(\alpha) = 0$

- (i) Prove that
- d
- is a multiple of
- α

2

- (ii) Prove that the polynomial
- $q(x) = 5x^4 - x^3 + 3x^2 - 3$

does not have an integer root.

2

- (b) Let
- $P(x) = x^3 - 11x - 14$

Factorise $P(x)$ over the reals and hence find the three roots of $P(x) = 0$

3

- (c) Find the roots of
- $q(x) = x^4 - 6x^3 + 12x^2 - 10x + 3$

given that it has a root of multiplicity 3.

2

- (d) Let
- α, β
- and
- γ
- be the roots of the equation
- $x^3 - 5x^2 + 5 = 0$

- (i) Find the value of
- $(\alpha-1)(\beta-1)(\gamma-1)$

2

- (ii) Find the value of
- $\alpha^3 + \beta^3 + \gamma^3$

2

- (iii) Find a polynomial equation with integer coefficients whose roots are

$$\alpha^2, \beta^2 \text{ and } \gamma^2$$

2

End of Question 5

Question 6 (15 marks) [START A NEW PAGE]

- (a) For the ellipse
- $\frac{x^2}{36} + \frac{y^2}{9} = 1$
- , find

- (i) the lengths of the axes

1

- (ii) the eccentricity

1

- (iii) the co-ordinates of the foci

1

- (iv) the equations of the directrices

1

- (b) Let
- $P(2 \sec \theta, \sqrt{5} \tan \theta)$
- be a variable point on the hyperbola
- $5x^2 - 4y^2 = 20$

The tangent at P meets the directrix at T . Show that PT subtends a right angle at the corresponding focus.

6

- (c) (i) If the line
- $y = mx + b$
- is a tangent to the ellipse
- $\frac{x^2}{6} + \frac{y^2}{3} = 1$
- ,

show that $b^2 = 6m^2 + 3$

2

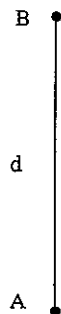
- (ii) The tangents to this ellipse from a point
- $P(X, Y)$
- meet at right angles. Prove that the locus of
- P
- is the circle
- $x^2 + y^2 = 9$
- .

3

End of Question 6

Question 7 (15 marks) [START A NEW PAGE]

Marks



A and B are two points d units apart in a vertical line. B is directly above A. Two identical particles are projected from A and B towards each other with the same velocity, u .

The resistance of the medium is kv per unit mass.

- (i) Draw a diagram indicating all forces acting on the particles. 1
- (ii) Consider the particle moving upward from A. By writing an expression for $\frac{dv}{dt}$,
- (α) show that $t = \frac{1}{k} \ln \left(\frac{g + ku}{g + kv} \right)$ 2
- (β) Hence, find v in terms of t 2
- (γ) Hence, find x in terms of t 2
- (iii) Consider the particle moving downward from B. Given that $\frac{dv}{dt} = g - kv$,
- (α) find t in terms of v . 2
- (β) Find v in terms of t 2
- (γ) Find x in terms of t . 2
- (iv) Hence, prove that the particles meet after a time of $\frac{1}{k} \ln \left(\frac{2u}{2u - kd} \right)$ 2

End of Question 7

Question 8 (15 marks) [START A NEW PAGE]

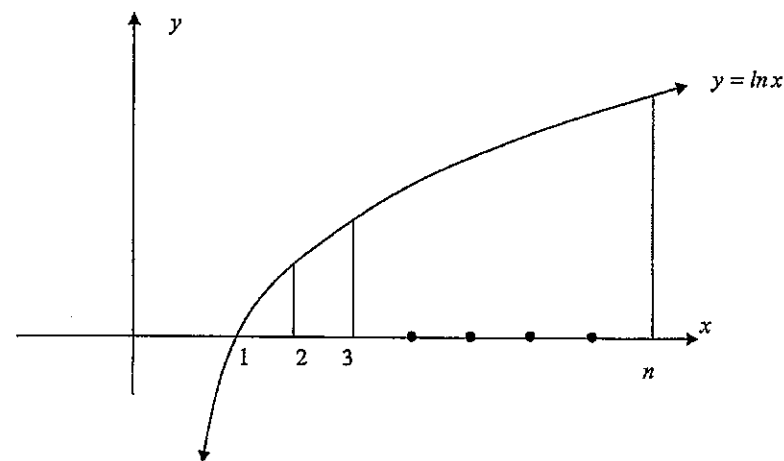
Marks

- (a) A particle is moving along the x axis. Its acceleration is given by

$$\frac{d^2 x}{dt^2} = \frac{12 - 4x}{x^3}. \text{ The particle starts from rest at the point } x = 6.$$

- (i) Show that the particle starts moving in the negative x direction. 1
- (ii) Find an expression for velocity, v , in terms of x . 3
- (iii) The path along which the particle moves is bounded. What part of the x axis is the path of the particle? 1

- (b) Consider the area under the curve $y = \ln x$ between $x = 1$ and $x = n$.



- (i) Show that this area is exactly equal to $\ln \left(\frac{n^n}{e^{n-1}} \right)$ 2

Question 8 continues on page 12

Marks

Question 8 (continued)

- (ii) Use the Trapezoidal Rule to find an expression which approximates this area.

2

- (iii) Hence show that $n^n > \sqrt{n} (n-1)! e^{n-1}$

1

- (c) Given that $\sin^{-1} 2x$, $\cos^{-1} 2x$ and $\sin^{-1} (1-2x)$ are all acute,

- (i) Show that $\sin [\cos^{-1} 2x - \sin^{-1} 2x] = 1-8x^2$

3

- (ii) Solve the equation $\cos^{-1} 2x - \sin^{-1} 2x = \sin^{-1} (1-2x)$

2

End of Paper

Question 1:

(a) $(\sqrt{3}+1)z = 4\sqrt{3}-4i$

(i) $z = \frac{4\sqrt{3}-4i}{\sqrt{3}+i} \times \frac{\sqrt{3}-i}{\sqrt{3}-i}$

$= \frac{12-4\sqrt{3}i-4\sqrt{3}i-4}{4}$

$= 2-2\sqrt{3}i$

(ii) $|z| = \sqrt{2^2 + (2\sqrt{3})^2}$

$= \sqrt{16}$

$= 4$

(iii) $\arg(z) = -\frac{\pi}{3}$

(iv) $z^8 = \left[4 \operatorname{cis} \left(-\frac{\pi}{3}\right)\right]^8$

$= 2^{16} \operatorname{cis} \left(-\frac{8\pi}{3}\right)$

$= 2^{16} \left[\cos \left(-\frac{2\pi}{3}\right) + i \sin \left(-\frac{2\pi}{3}\right) \right]$

$= 2^{16} \left\{ -\frac{1}{2} - \frac{\sqrt{3}}{2}i \right\}$

$= -2^{15} - 2^{15}\sqrt{3}i$

1 (b) Let $a + bi = \sqrt{-5 - 12i}$ then

$$a^2 - b^2 = -5 \quad \text{and} \quad 2abi = -12i$$
$$b = \frac{-6}{a}$$

$$\text{So } a^2 - \left(\frac{-6}{a}\right)^2 = -5$$

$$a^4 + 5a^2 - 36 = 0$$

$$(a^2 - 4)(a^2 + 9) = 0$$

$$\therefore a = \pm 2$$

$$\text{When } a = 2, \quad b = -3.$$

$$\text{When } a = -2, \quad b = 3.$$

$$\text{Hence } \sqrt{-5 - 12i} = \pm(2 - 3i).$$

Solving $z^2 + (1 - 2i)z + \left(\frac{1}{2} + 2i\right) = 0$ we have

$$z = \frac{-(1 - 2i) \pm \sqrt{(1 - 2i)^2 - 4\left(\frac{1}{2} + 2i\right)}}{2}$$

$$z = \frac{-1 + 2i \pm \sqrt{-5 - 12i}}{2}$$

$$z = \frac{-1 + 2i \pm (2 - 3i)}{2}$$

$$z = \frac{1 - i}{2} \quad \text{and} \quad z = \frac{-3 + 5i}{2}$$

1 (c) If w is a complex cube root of unity, then

$$w^3 - 1 = 0$$

$$\text{ie, } (w - 1)(w^2 + w + 1) = 0.$$

But $w \neq 1$, hence $1 + w + w^2 = 0$, as required.

RTP that

$$(1 + w)(1 + 2w)(1 + 3w)(1 + 7w) = 31 + 2w.$$

$$\text{LHS} = (1 + 3w + 2w^2)(1 + 10w + 21w^2)$$

$$= 1 + 10w + 21w^2 + 3w + 30w^2 + 63w^3 + 2w^2 + 20w^3 + 42w^4$$

$$= 84 + 13w + 53w^2 + 42w^4 \quad (\text{because } w^3 = 1)$$

$$= 84 + 55w + 53w^2 \quad (\text{because } w^4 = w)$$

$$= (53 + 53w + 53w^2) + (31 + 2w)$$

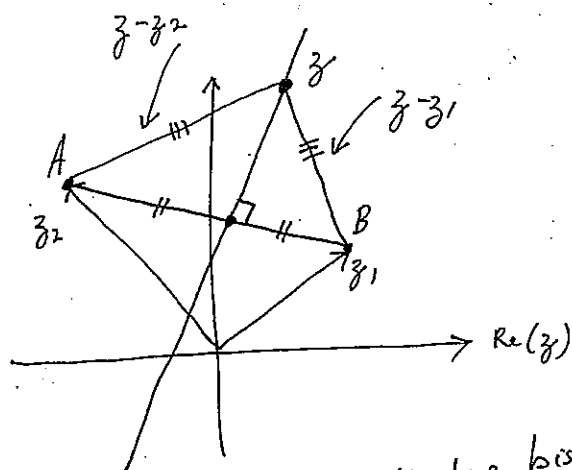
$$= 53(1 + w + w^2) + (31 + 2w)$$

$$= 31 + 2w \quad (\text{because } 1 + w + w^2 = 0)$$

$$= \text{RHS.}$$

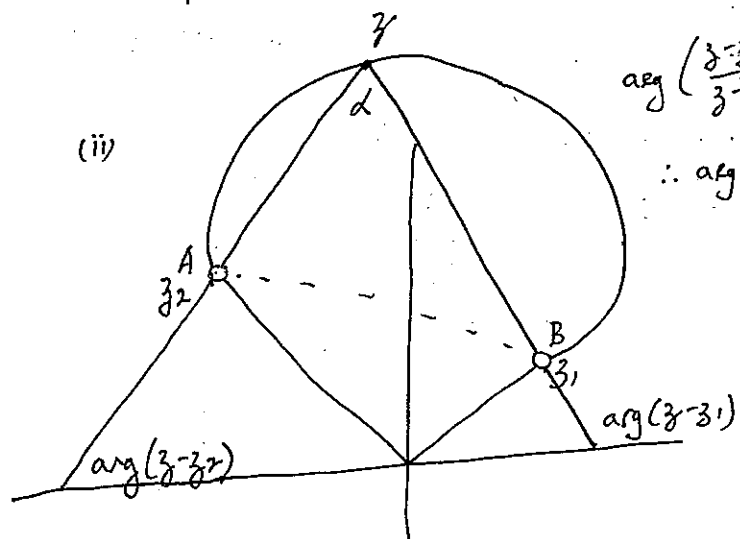
1 Cd)

(i)



The locus of z is the perpendicular bisector of AB .

(ii)

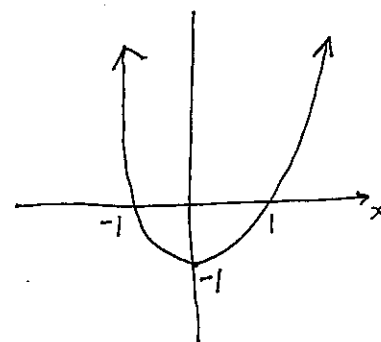


The locus of z is the upper part of the circle above the chord AB .

pg 4

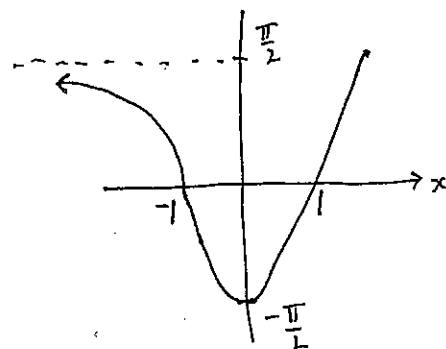
Question 2: (a)

(i) $y = f(|x|)$



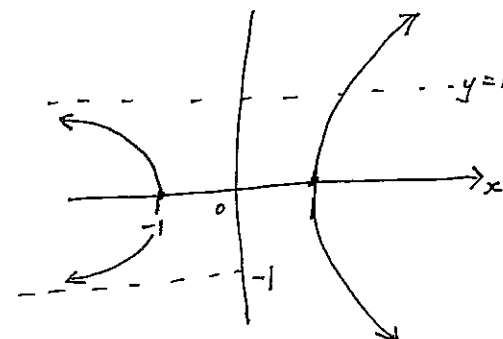
The left branch is the reflection of the right branch in the y axis.

(iii) $y = \sin^{-1}[f(x)]$

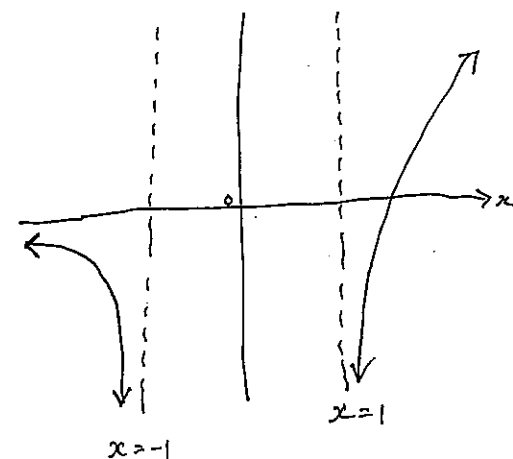


(ii) $y^2 = f(x)$

i.e. $y = \pm \sqrt{f(x)}$



(iv) $y = \ln[f(x)]$



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2(b)

(i) Suppose that $x=0$ were in the domain.
Then, by substitution, into the equation of the curve, we would have

$$0 \times y^2 + 0^2 = 1$$

i.e. $0 = 1$, which is false.

Hence $x=0$ is not in the domain.

(ii) Substitute $y=0$ and solve $x^2=1$.

The x intercepts are $(1, 0)$ and $(-1, 0)$.

$$(iii) \quad y^2 = \frac{1-x^2}{x}$$

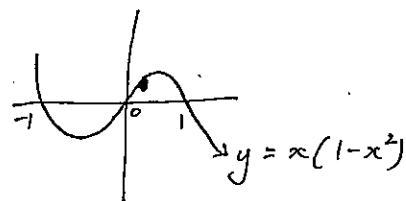
$$y = \pm \sqrt{\frac{1-x^2}{x}}$$

$$\text{Need } \frac{1-x^2}{x} \geq 0$$

$$\text{i.e. } x(1-x^2) \geq 0$$

$\therefore$ The domain is

$$x \leq -1 \quad \text{or} \quad 0 < x \leq 1.$$



2b

(iv)

$$y = \pm \sqrt{\frac{1-x^2}{x}}$$

$$\text{i.e. } y = \pm \sqrt{\frac{1}{x} - x}$$

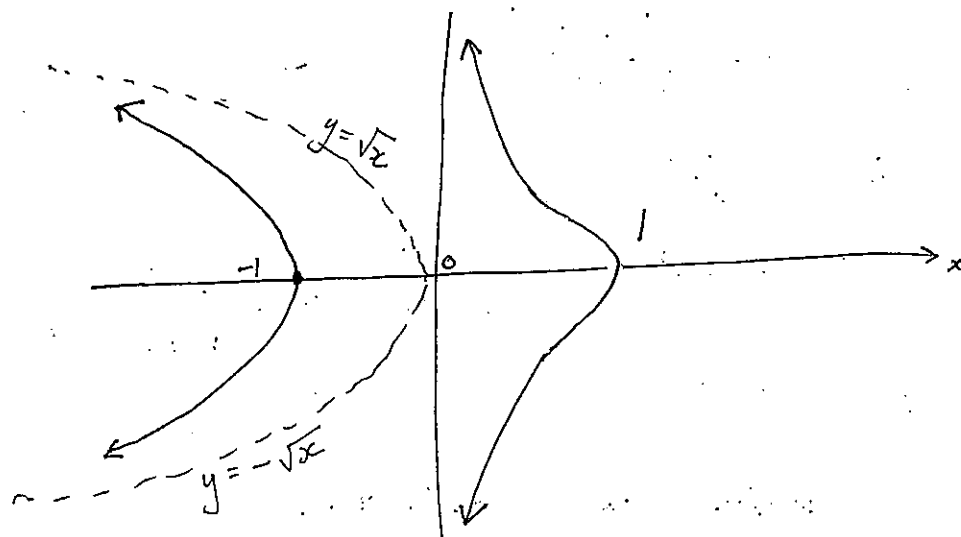
$$\text{as } x \rightarrow -\infty, \frac{1}{x} \rightarrow 0$$

$$\text{Hence } y \rightarrow \pm \sqrt{-x}. \quad [y^2 \rightarrow -x]$$

Hence the asymptotes are $x=0$ and $y = \pm \sqrt{-x}$

$$\text{i.e. } x=0 \text{ and } y^2 = -x$$

$$(v) \quad xy^2 + x^2 = 1$$



Question 3: (a)

$$(i) \int \frac{(x+3) dx}{\sqrt[3]{x^2+6x}}$$

$$\text{Let } u = x^2 + 6x$$

$$du = (2x+6) dx$$

$$\frac{1}{2} du = (x+3) dx$$

$$I = \int \frac{\frac{1}{2} du}{u^{1/3}}$$

$$= \frac{1}{2} \int u^{-1/3} du$$

$$= \frac{3}{4} u^{2/3}$$

$$= \frac{3}{4} (x^2+6x)^{2/3} + C.$$

$$(ii) \int \frac{dx}{2 + \cos x}$$

$$\text{Let } t = \tan\left(\frac{x}{2}\right)$$

$$\frac{dt}{dx} = \frac{1}{2} \sec^2\left(\frac{x}{2}\right)$$

$$= \frac{1}{2} (1+t^2)$$

$$\therefore dx = \frac{2 dt}{1+t^2}$$

$$\text{And } 2 + \cos x = 2 + \frac{1-t^2}{1+t^2}$$

$$= \frac{2+2t^2+1-t^2}{1+t^2} = \frac{3+t^2}{1+t^2}$$

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$$(ii) \int \frac{dy}{y^2 + 10y + 30}$$

$$= \int \frac{dy}{5 + (y+5)^2}$$

$$= \frac{1}{\sqrt{5}} \tan^{-1}\left(\frac{y+5}{\sqrt{5}}\right) + C.$$

$$\text{Hence } I = \int \frac{\frac{2dt}{1+t^2}}{\frac{3+t^2}{1+t^2}}$$

$$= \int \frac{2 dt}{3+t^2}$$

$$= \frac{2}{\sqrt{3}} \tan^{-1}\left(\frac{t}{\sqrt{3}}\right)$$

$$= \frac{2}{\sqrt{3}} \tan^{-1}\left(\frac{\tan(x/2)}{\sqrt{3}}\right) + C.$$

3(a)

$$(iv) \int x^2 e^{2x} dx$$

$$= \frac{1}{2} e^{2x} \cdot x^2 - \int \frac{1}{2} e^{2x} \cdot (2x) dx$$

$$= \frac{1}{2} x^2 e^{2x} - \int x e^{2x} dx$$

$$= \frac{1}{2} x^2 e^{2x} - \left\{ \frac{1}{2} e^{2x} \cdot x - \int \frac{1}{2} e^{2x} (1) dx \right\}$$

$$= \frac{1}{2} x^2 e^{2x} - \frac{1}{2} x e^{2x} + \frac{1}{2} \int e^{2x} dx$$

$$= \frac{1}{2} x^2 e^{2x} - \frac{1}{2} x e^{2x} + \frac{1}{4} e^{2x} + C.$$

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$$3(b) \quad x^3 + x^2 - 6x$$

$$= x(x^2 + x - 6)$$

$$= x(x+3)(x-2).$$

$$\text{let } \frac{x+1}{x^3+x^2-6x} = \frac{A}{x} + \frac{B}{x-2} + \frac{C}{x+3}, \text{ then}$$

$$x+1 = A(x+3)(x-2) + Bx(x+3) + Cx(x-2).$$

$$\text{let } x=0. \text{ Then } 1 = -6A \Rightarrow A = -\frac{1}{6}.$$

$$\text{let } x=2. \text{ Then } 3 = 10B \Rightarrow B = \frac{3}{10}.$$

$$\text{let } x=-3. \text{ Then } -2 = 15C \Rightarrow C = -\frac{2}{15}.$$

$$\text{Hence } \int \frac{(x+1) dx}{x^3+x^2-6x}$$

$$= \int \left(\frac{-1/6}{x} + \frac{3/10}{x-2} - \frac{2/15}{x+3} \right) dx$$

$$= -\frac{1}{6} \ln(x) + \frac{3}{10} \ln(x-2) - \frac{2}{15} \ln(x+3) + C.$$

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Question 4:

$$(a) \quad I_n = \int_0^{\pi/4} \tan^n x \, dx$$

$$= \int_0^{\pi/4} \tan^{n-2} x \cdot (\sec^2 x - 1) \, dx$$

$$= \int_0^{\pi/4} \tan^{n-2} x \cdot \sec^2 x \, dx - \int_0^{\pi/4} \tan^{n-2} x \, dx$$

$$= \left[\frac{1}{n-1} \tan^{n-1} x \right]_0^{\pi/4} - I_{n-2}$$

$$\therefore I_n + I_{n-2} = \frac{1}{n-1} \left(\tan \frac{\pi}{4} \right)^{n-1}$$

$$\therefore I_n + I_{n-2} = \frac{1}{n-1}, \text{ as required.}$$

$$I_5 + I_3 = \frac{1}{4}$$

$$I_3 + I_1 = \frac{1}{2}$$

$$I_1 = \int_0^{\pi/4} \tan x \, dx$$

$$= \left[-\ln(\cos x) \right]_0^{\pi/4}$$

$$= -\ln\left(\frac{1}{\sqrt{2}}\right)$$

$$\therefore I_3 = \frac{1}{2} + \ln\left(\frac{1}{\sqrt{2}}\right)$$

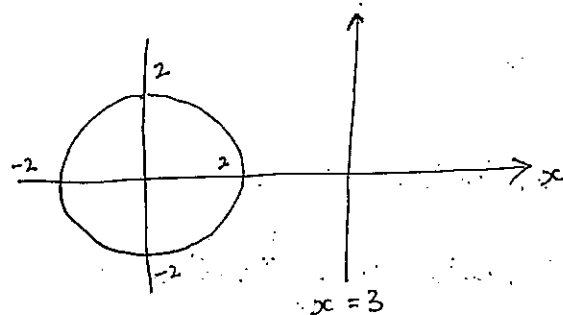
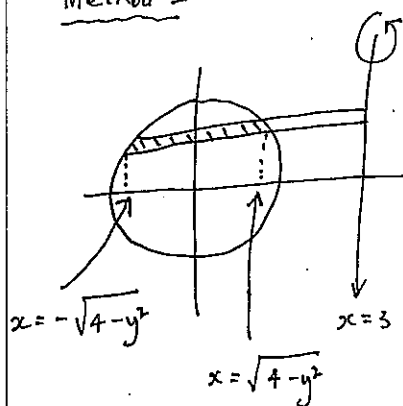
$$\therefore I_5 = \frac{1}{4} - \left(\frac{1}{2} + \ln\left(\frac{1}{\sqrt{2}}\right) \right)$$

$$\therefore I_5 = -\frac{1}{4} - \ln\left(\frac{1}{\sqrt{2}}\right)$$

$$\therefore I_5 = \ln \sqrt{2} - \frac{1}{4}$$

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4(b)

Method 1:

Rotation of the shaded strip about the line $x=3$ produces a "washer" of Volume ΔV , where

$$\Delta V = \pi (R^2 - r^2) \Delta y$$

$$\text{and } R = 3 + \sqrt{4-y^2}$$

$$\text{and } r = 3 - \sqrt{4-y^2}$$

$$\text{Hence } \Delta V = \pi (R^2 - r^2) \Delta y$$

$$= \pi \{ (3 + \sqrt{4-y^2})^2 - (3 - \sqrt{4-y^2})^2 \} \Delta y$$

$$= \pi \{ 6(2\sqrt{4-y^2}) \} \Delta y$$

$$= 12\pi \sqrt{4-y^2} \Delta y$$

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4(b)

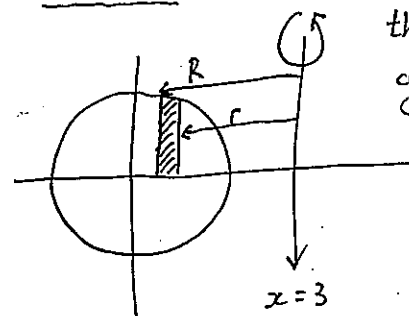
Hence the required Volume is V , where

$$V = 12\pi \int_{-2}^2 \sqrt{4-y^2} dy$$

$$= 24\pi \int_0^2 \sqrt{4-y^2} dy$$

$$= 24\pi \left(\frac{1}{4} \pi \times 2^2 \right)$$

$$= 24\pi^2 \text{ units}^3$$

Method 2:

Rotation of the shaded strip about the line $x=3$ produces a cylindrical shell of Volume ΔV , where

$$\Delta V = \pi (R^2 - r^2) \times \text{height}$$

$$\text{Now } R = (3-x)$$

$$r = (3 - (x + \Delta x))$$

$$\text{height} = \sqrt{4-x^2}$$

$$\text{Hence } \Delta V = \pi \{ (3-x)^2 - (3-(x+\Delta x))^2 \} \times \sqrt{4-x^2}$$

$$= \pi \{ (6-2x-\Delta x)(\Delta x) \} \sqrt{4-x^2}$$

$$= 2\pi (3-x) \sqrt{4-x^2} \Delta x$$

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Hence the Required volume is V , where

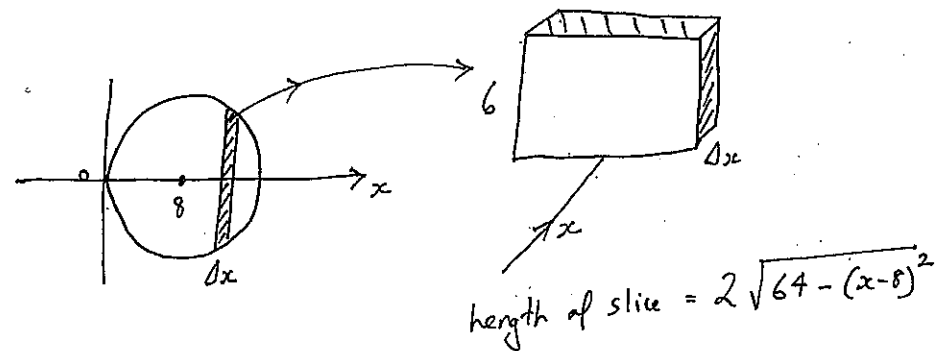
$$\begin{aligned}
 V &= 2 \times 2\pi \int_{-2}^2 (3-x) \sqrt{4-x^2} \, dx \\
 &= 4\pi \int_{-2}^2 3\sqrt{4-x^2} \, dx - 4\pi \int_{-2}^2 x\sqrt{4-x^2} \, dx \\
 &= 12\pi \left(\frac{1}{2} \times \pi \times 2^2 \right) - 4\pi \left[\frac{-1}{2} \cdot \frac{2}{3} (4-x^2)^{\frac{3}{2}} \right]_{-2}^2 \\
 &= 24\pi^2 - 0 \\
 &= 24\pi^2 \text{ units}^3
 \end{aligned}$$

4(c) (i) $\frac{d}{dx} \left\{ \frac{1}{2} x (a^2 - x^2)^{\frac{1}{2}} + \frac{1}{2} a^2 \sin^{-1} \left(\frac{x}{a} \right) \right\}$

$$\begin{aligned}
 &= \frac{1}{2} x \cdot \frac{1}{2} (a^2 - x^2)^{-\frac{1}{2}} (-2x) + \frac{1}{2} (a^2 - x^2)^{\frac{1}{2}} + \\
 &\quad \frac{1}{2} a^2 \cdot \frac{1}{\sqrt{1 - \frac{x^2}{a^2}}} \cdot \frac{1}{a} \\
 &= \frac{-x^2}{2\sqrt{a^2 - x^2}} + \frac{\sqrt{a^2 - x^2}}{2} + \frac{a^2}{2\sqrt{a^2 - x^2}} \\
 &= \frac{a^2 - x^2}{2\sqrt{a^2 - x^2}} + \frac{\sqrt{a^2 - x^2}}{2} \\
 &= \frac{a^2 - x^2}{2\sqrt{a^2 - x^2}} + \frac{a^2 - x^2}{2\sqrt{a^2 - x^2}} = \frac{a^2 - x^2}{\sqrt{a^2 - x^2}} = \sqrt{a^2 - x^2}
 \end{aligned}$$

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4(c) (ii) $x^2 + y^2 = 16x$
 $x^2 - 16x + 64 + y^2 = 64$
 $(x-8)^2 + y^2 = 64$



The Volume of a slice is ΔV , where

$$\begin{aligned}
 \Delta V &= 6 \times 2\sqrt{64 - (x-8)^2} \times \Delta x \\
 &= 12\sqrt{64 - (x-8)^2} \Delta x
 \end{aligned}$$

Hence the Volume, V , of the Required solid is

$$\begin{aligned}
 V &= 12 \int_0^{16} \sqrt{64 - (x-8)^2} \, dx \\
 &= 12 \left\{ \left[\frac{1}{2} (x-8) \sqrt{64 - (x-8)^2} + \frac{1}{2} \times 64 \times \sin^{-1} \left(\frac{x-8}{8} \right) \right]_0^{16} \right\} \\
 &= 12 \left\{ (0 + 32(\frac{\pi}{2})) - (0 + 32(-\frac{\pi}{2})) \right\} \\
 &= 12 \times 32\pi \\
 &= 384\pi \text{ units}^3
 \end{aligned}$$

Question 5:

(a) (i) $p(x) = ax^4 + bx^3 + cx^2 + d$

$$p(d) = ad^4 + bd^3 + cd^2 + d = 0$$

$$\therefore d = -ad^4 - bd^3 - cd^2$$

$$\therefore d = -d(ad^3 + bd^2 + cd)$$

which is divisible by d .

Hence d is a multiple of d .

(ii) Using the result from part (i), if $q(x)$ has an integer root, then that integer divides -3 .
The divisors of -3 are $\pm 1, \pm 3$.

Now $q(1) = 5 - 1 + 3 - 3 \neq 0$.

And $q(-1) = 5 + 1 + 3 - 3 \neq 0$.

And $q(3) = 405 - 27 + 27 - 3 \neq 0$.

And $q(-3) = 405 + 27 + 27 - 3 \neq 0$.

Hence $q(x)$ does not have an integer root.

5(b) $P(x) = x^3 - 11x - 14$

Notice that $P(-2) = -8 + 22 - 14 = 0$.

Hence $(x+2)$ is a factor of $P(x)$.

$$\begin{array}{r} x^2 - 2x - 7 \\ x+2 \overline{) x^3 + 0x^2 - 11x - 14} \\ \underline{x^3 + 2x^2} \\ -2x^2 - 11x - 14 \\ \underline{-2x^2 - 4x} \\ -7x - 14 \\ \underline{-7x - 14} \\ 0 \end{array}$$

$$\therefore P(x) = (x+2)(x^2 - 2x - 7)$$

Solving $x^2 - 2x - 7 = 0$

$$x = \frac{2 \pm \sqrt{4 - 4(-7)}}{2}$$

$$x = \frac{2 \pm \sqrt{32}}{2} = \frac{2 \pm 4\sqrt{2}}{2} = 1 \pm 2\sqrt{2}$$

The three roots of $P(x) = 0$ are

$$x = -2 \text{ and } 1 \pm 2\sqrt{2}$$

$$5(c) \quad q(x) = x^4 - 6x^3 + 12x^2 - 10x + 3$$

$$q'(x) = 4x^3 - 18x^2 + 24x - 10$$

$$\begin{aligned} q''(x) &= 12x^2 - 36x + 24 \\ &= 12(x^2 - 3x + 2) \\ &= 12(x-1)(x-2) \end{aligned}$$

Test $x=1$:

$$q'(1) = 4 - 18 + 24 - 10 = 0$$

$$q(1) = 1 - 6 + 12 - 10 + 3 = 0.$$

Hence $x=1$ is a root of multiplicity 3.

$$\text{Hence } q(x) = (x-1)^3(x-3).$$

So the roots of $q(x)=0$ are $x=1, 1, 1, 3$.

$$(d) \quad x^3 - 5x^2 + 5 = 0 \quad \begin{cases} \alpha + \beta = 5 - \gamma \\ \alpha\beta + \alpha\gamma + \beta\gamma = 0 \\ \alpha\beta\gamma = -5 \end{cases}$$

$$(i) \quad (\alpha-1)(\beta-1)(\gamma-1)$$

$$= \alpha\beta\gamma - (\alpha\beta + \alpha\gamma + \beta\gamma) + (\alpha + \beta + \gamma) - 1$$

$$= -5 + 5 - 1$$

$$= -1$$

$$5(d) (ii) \quad x^3 - 5x^2 + 5 = 0$$

$$\text{So } \alpha^3 - 5\alpha^2 + 5 = 0$$

$$\text{and } \beta^3 - 5\beta^2 + 5 = 0$$

$$\text{and } \gamma^3 - 5\gamma^2 + 5 = 0$$

$$\therefore \alpha^3 + \beta^3 + \gamma^3 = 5(\alpha^2 + \beta^2 + \gamma^2) - 15.$$

$$\begin{aligned} \text{Now } \alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) \\ &= (5)^2 - 2(0) \\ &= 25. \end{aligned}$$

$$\therefore \alpha^3 + \beta^3 + \gamma^3 = 5 \times 25 - 15 = 110.$$

(iii) The equation whose roots are α^2, β^2 and γ^2 is

$$(\sqrt{x})^3 - 5(\sqrt{x})^2 + 5 = 0$$

$$x\sqrt{x} - 5x + 5 = 0$$

$$x\sqrt{x} = 5x - 5$$

$$(x\sqrt{x})^2 = (5x - 5)^2$$

$$x^3 = 25x^2 - 50x + 25$$

$$\text{ie, } x^3 - 25x^2 + 50x - 25 = 0.$$

Question 6:

(a) $\frac{x^2}{36} + \frac{y^2}{9} = 1 \Rightarrow a=6, b=3.$

(i) Length of major axis is 12 units.
Length of minor axis is 6 units.

(ii) $b^2 = a^2(1-e^2)$

$9 = 36(1-e^2)$

$e^2 = \frac{3}{4} \Rightarrow e = \frac{\sqrt{3}}{2}$

(iii) The foci are at $(\pm ae, 0)$.

The foci are at $(\pm 3\sqrt{3}, 0)$.

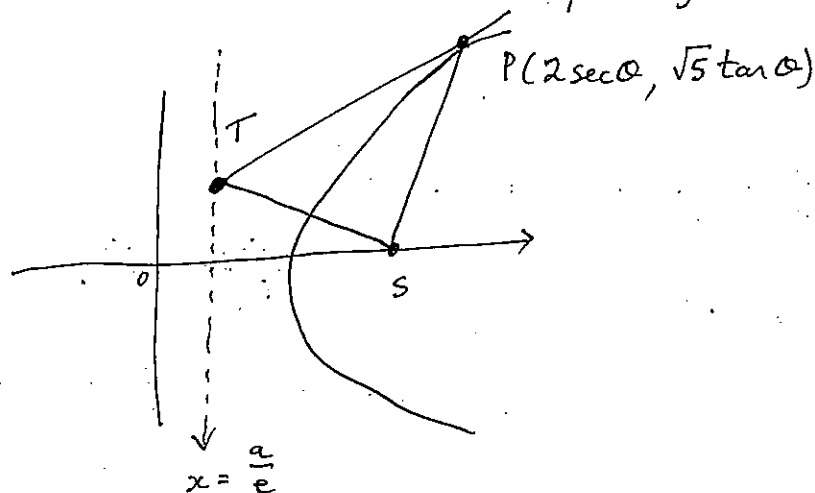
(iv) The directrices have equations $x = \pm \frac{a}{e}$.

The directrices have equations $x = \pm \frac{6}{\sqrt{3}/2} = \pm \frac{12}{\sqrt{3}}$

ie, $x = \pm 4\sqrt{3}.$

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6(b) $5x^2 - 4y^2 = 20$, ie, $\frac{x^2}{4} - \frac{y^2}{5} = 1.$



To find the equation of the tangent at P:

$$\frac{2x}{4} - \frac{2y}{5} \cdot \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{2x}{4} \cdot \frac{5}{2y} = \frac{5x}{4y}$$

At $P(2\sec\theta, \sqrt{5}\tan\theta)$, $\frac{dy}{dx} = \frac{10\sec\theta}{4\sqrt{5}\tan\theta}$

Hence the equation of the tangent at P is

$$y - \sqrt{5}\tan\theta = \frac{10\sec\theta}{4\sqrt{5}\tan\theta} (x - 2\sec\theta), \text{ etc.}$$

..... ①

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6(b) ctd.

To find the equation of the directrix, first find the eccentricity:

$$5 = 4(e^2 - 1)$$

$$\frac{5}{4} + \frac{4}{4} = \frac{9}{4} = e^2 \Rightarrow e = \frac{3}{2}$$

Hence the directrix has equation $x = \frac{4}{3/2} = \frac{4}{3}$.

To find the y-coordinate of T: substitute $x = \frac{4}{3}$ in (1):

$$y - \sqrt{5} \tan \theta = \frac{10 \sec \theta}{4\sqrt{5} \tan \theta} \left(\frac{4}{3} - 2 \sec \theta \right)$$

$$\therefore y = \frac{10 \sec \theta}{3\sqrt{5} \tan \theta} - \frac{10 \sec^2 \theta}{2\sqrt{5} \tan \theta} + \sqrt{5} \tan \theta$$

$$\therefore y = \frac{2\sqrt{5}}{3 \sin \theta} - \frac{\sqrt{5}(1 + \tan^2 \theta)}{\tan \theta} + \sqrt{5} \tan \theta$$

$$\therefore y = \frac{2\sqrt{5}}{3 \sin \theta} - \frac{\sqrt{5} \cos \theta}{\sin \theta} = \frac{2\sqrt{5} - 3\sqrt{5} \cos \theta}{3 \sin \theta}$$

$$\text{Hence } T = \left\{ \frac{4}{3}, \frac{2\sqrt{5} - 3\sqrt{5} \cos \theta}{3 \sin \theta} \right\}$$

$$\text{And } S = \{3, 0\}.$$

6(c)

(i) Substitute $y = mx + b$ into the equation of ellipse:

$$\frac{x^2}{6} + \frac{(mx+b)^2}{3} = 1$$

$$\text{i.e., } x^2 + 2(m^2x^2 + 2bmx + b^2) = 6$$

$$\text{i.e., } x^2(2m^2+1) + 4bmx + (2b^2-6) = 0.$$

We need $\Delta = 0$, i.e.,

$$(4bm)^2 - 4(2m^2+1)(2b^2-6) = 0$$

$$16b^2m^2 - 4(4b^2m^2 - 12m^2 + 2b^2 - 6) = 0$$

$$16b^2m^2 - 16b^2m^2 + 48m^2 - 8b^2 + 24 = 0$$

$$\text{i.e., } 12m^2 - 2b^2 + 6 = 0$$

$$\text{i.e., } 6m^2 - b^2 + 3 = 0$$

$$\text{i.e., } b^2 = 6m^2 + 3, \text{ as required.}$$

6C(i) std
Hence m of PS = $\frac{\sqrt{5} \tan \theta}{2 \sec \theta - 3}$

and m of ST = $\frac{2\sqrt{5} - 3\sqrt{5} \cos \theta}{3 \sin \theta}$

$\frac{4}{3} - 3$

= $\frac{2\sqrt{5} - 3\sqrt{5} \cos \theta}{-5 \sin \theta}$

= $\frac{3\sqrt{5} \cos \theta - 2\sqrt{5}}{5 \sin \theta}$

Now $(m \text{ of PS}) \times (m \text{ of ST})$

= $\frac{\sqrt{5} \tan \theta}{2 \sec \theta - 3} \times \frac{3\sqrt{5} \cos \theta - 2\sqrt{5}}{5 \sin \theta}$

= $\frac{15 \sin \theta - 10 \tan \theta}{10 \tan \theta - 15 \sin \theta}$

= $\frac{-(10 \tan \theta - 15 \sin \theta)}{(10 \tan \theta - 15 \sin \theta)}$

= -1.

Hence $\angle PST = 90^\circ$, as required.

6C(ii) The tangent $y = mx + b$ passes through (X, Y) so

$Y = mX + b$

ie, $b = (Y - mX)$

ie, $b^2 = Y^2 - 2mXY + m^2X^2$

But $b^2 = 6m^2 + 3$, so

$6m^2 + 3 = Y^2 - 2mXY + m^2X^2$

ie, $0 = m^2X^2 - 6m^2 - 2mXY + Y^2 - 3$

ie, $0 = m^2(X^2 - 6) - 2mXY + (Y^2 - 3)$

Now this is a quadratic in m with two roots, namely m and $-\frac{1}{m}$. Hence the product of the roots is -1. But the product of the roots is also equal to $\frac{c}{a}$, ie, $\frac{Y^2 - 3}{X^2 - 6}$

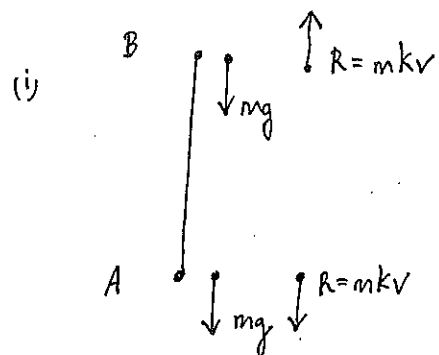
Hence $\frac{Y^2 - 3}{X^2 - 6} = -1$

ie, $Y^2 - 3 = 6 - X^2$

ie, $X^2 + Y^2 = 9$

ie, the locus of $P(X, Y)$ is the circle $x^2 + y^2 = 9$.

Question 7:



(ii) $m\ddot{x} = -mg - mkv$
 $\ddot{x} = -g - kv = \frac{dv}{dt}$

(d) $\frac{dt}{dv} = \frac{-1}{g+kv}$

$t = \int \frac{-dv}{g+kv}$

$t = -\frac{1}{k} \ln(g+kv) + C$

But when $t=0$, $v=u$ so $C = \frac{1}{k} \ln(g+ku)$.

Hence $t = -\frac{1}{k} \ln(g+kv) + \frac{1}{k} \ln(g+ku)$

i.e., $t = \frac{1}{k} \ln\left(\frac{g+ku}{g+kv}\right)$, as required.

(b) So $e^{kt} = \left(\frac{g+ku}{g+kv}\right)$

$g+kv = (g+ku)e^{-kt}$

So $v = \frac{1}{k}(g+ku)e^{-kt} - \frac{g}{k}$

7(ii)

(8) So $\frac{dx}{dt} = \frac{1}{k}(g+ku)e^{-kt} - \frac{g}{k}$

$\therefore x = -\frac{1}{k^2}(g+ku)e^{-kt} - \frac{gt}{k} + F$

But when $t=0$, $x=0$ so $F = \frac{1}{k^2}(g+ku)$

$\therefore x = -\frac{1}{k^2}(g+ku)e^{-kt} - \frac{gt}{k} + \frac{1}{k^2}(g+ku)$

7(iii) $\frac{dv}{dt} = g - kv$

(d) $\frac{dt}{dv} = \frac{1}{g-kv}$

$\therefore t = \int \frac{dv}{g-kv} = -\frac{1}{k} \ln(g-kv) + H$

But when $t=0$, $v=u$ so $H = \frac{1}{k} \ln(g-ku)$

$\therefore t = -\frac{1}{k} \ln(g-kv) + \frac{1}{k} \ln(g-ku)$

i.e., $t = \frac{1}{k} \ln\left(\frac{g-ku}{g-kv}\right)$

(b) So $e^{kt} = \left(\frac{g-ku}{g-kv}\right)$

$g-kv = (g-ku)e^{-kt}$

$$\therefore kv = g - (g - ku)e^{-kt}$$

$$\therefore v = \frac{g}{k} - \frac{1}{k}(g - ku)e^{-kt}$$

7(iii)
(8) $\frac{dx}{dt} = \frac{g}{k} - \frac{1}{k}(g - ku)e^{-kt}$

$$\therefore x = \frac{gt}{k} + \frac{1}{k^2}(g - ku)e^{-kt} + R$$

But when $t=0$, $x=0$ so $R = -\frac{1}{k^2}(g - ku)$

$$\therefore x = \frac{1}{k^2}(g - ku)e^{-kt} + \frac{gt}{k} - \frac{1}{k^2}(g - ku)$$

(iv) Let the particles meet after t units of time.
A will have travelled a distance of x_1 , and
B will have travelled a distance of x_2 such that

$$x_1 + x_2 = d, \text{ i.e.,}$$

$$\left\{ -\frac{1}{k^2}(g + ku)e^{-kt} - \frac{gt}{k} + \frac{1}{k^2}(g + ku) + \frac{1}{k^2}(g - ku)e^{-kt} + \frac{gt}{k} - \frac{1}{k^2}(g - ku) \right\} = d$$

i.e., $\frac{2u}{k} - \frac{2u}{k}e^{-kt} = d$

i.e., $\frac{k d}{2u} = 1 - e^{-kt}$ i.e., $e^{-kt} = \frac{2u - kd}{2u}$

i.e., $e^{kt} = \frac{2u}{2u - kd}$ i.e., $t = \frac{1}{k} \ln \left(\frac{2u}{2u - kd} \right)$,
as required.

Question 8: (a)

(ii) $\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = \frac{12 - 4x}{x^3} = \frac{12}{x^3} - \frac{4}{x^2}$

$$\therefore \frac{1}{2} v^2 = -\frac{6}{x^2} + \frac{4}{x} + C$$

But when $x=6$, $v=0$ so

$$0 = -\frac{6}{36} + \frac{4}{6} + C \Rightarrow C = -\frac{1}{2}$$

$$\therefore v^2 = -\frac{12}{x^2} + \frac{8}{x} - 1$$

$$\therefore v^2 = \frac{8x - x^2 - 12}{x^2}$$

$$\therefore v = \pm \sqrt{\frac{8x - x^2 - 12}{x^2}} = \pm \frac{\sqrt{8x - x^2 - 12}}{x}$$

(i) When $x=6$, $v=0$ and $\frac{dv^2}{dx} = \frac{12 - 24}{64} < 0$.

Since acceleration is negative, and $v=0$, therefore it starts moving in the negative direction.

(iii) $v = \pm \frac{\sqrt{8x - x^2 - 12}}{x}$

For v to be real, we need $8x - x^2 - 12 \geq 0$, i.e.,

$$(x-2)(6-x) \geq 0,$$

i.e., $2 \leq x \leq 6$.

The particle is restricted to $2 \leq x \leq 6$.

8(b) (i) Exact area $= \int_1^n \log_e x \, dx$

$$= [x \ln x - x]_1^n$$

$$= (n \ln n - n) - (0 - 1)$$

$$= n \ln n - n + 1$$

$$= (\ln n^n) - (n-1)$$

$$= (\ln n^n) - \log_e e^{(n-1)}$$

$$= \ln \left(\frac{n^n}{e^{n-1}} \right), \text{ as required.}$$

(ii) Using the Trapezoidal Rule with strips of width 1 unit,

$$\text{Area} \doteq \frac{1}{2} \{ 0 + \ln n + 2[\ln 2 + \ln 3 + \dots + \ln(n-1)] \}$$

(iii) Area by Trapezoidal Rule < Exact area

$$\frac{1}{2} \ln n + \ln(n-1)! < \ln \left(\frac{n^n}{e^{n-1}} \right)$$

$$\ln \sqrt{n} (n-1)! < \ln \left(\frac{n^n}{e^{n-1}} \right)$$

$$\sqrt{n} (n-1)! < \left(\frac{n^n}{e^{n-1}} \right)$$

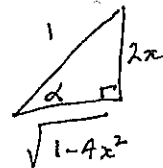
$$\sqrt{n} (n-1)! e^{n-1} < n^n$$

$$\therefore n^n > \sqrt{n} (n-1)! e^{n-1}, \text{ as required.}$$

8(c) (i) let $\alpha = \sin^{-1} 2x$

$$\sin \alpha = 2x$$

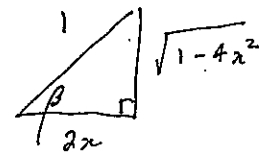
$$\cos \alpha = \sqrt{1-4x^2}$$



let $\beta = \cos^{-1} 2x$

$$\cos \beta = 2x$$

$$\sin \beta = \sqrt{1-4x^2}$$



Now $\sin [\cos^{-1} 2x - \sin^{-1} 2x]$

$$= \sin (\beta - \alpha)$$

$$= \sin \beta \cos \alpha - \cos \beta \sin \alpha$$

$$= \sqrt{1-4x^2} \times \sqrt{1-4x^2} - 2x \times 2x$$

$$= 1 - 4x^2 - 4x^2$$

$$= 1 - 8x^2, \text{ as required.}$$

(ii) To solve $\cos^{-1} 2x - \sin^{-1} 2x = \sin^{-1} (1-2x)$

$$\sin [\cos^{-1} 2x - \sin^{-1} 2x] = \sin [\sin^{-1} (1-2x)]$$

$$1 - 8x^2 = 1 - 2x, \text{ from part (i)}$$

$$\therefore 0 = 8x^2 - 2x$$

$$0 = 2x(4x-1)$$

$$\therefore x = 0 \text{ or } \frac{1}{4}$$

However $\sin^{-1} 2x, \cos^{-1} 2x$ & $\sin^{-1} (1-2x)$ are real, so $x = \frac{1}{4}$ only.